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(54) **ELECTRICAL CONNECTOR MOUNTING**

(71) Applicant: **GM GLOBAL TECHNOLOGY OPERATIONS LLC**, Detroit, MI (US)

(72) Inventor: **Thomas Welsh**, Plymouth, MI (US)

(73) Assignee: **GM GLOBAL TECHNOLOGY OPERATIONS LLC**, Detroit, MI (US)

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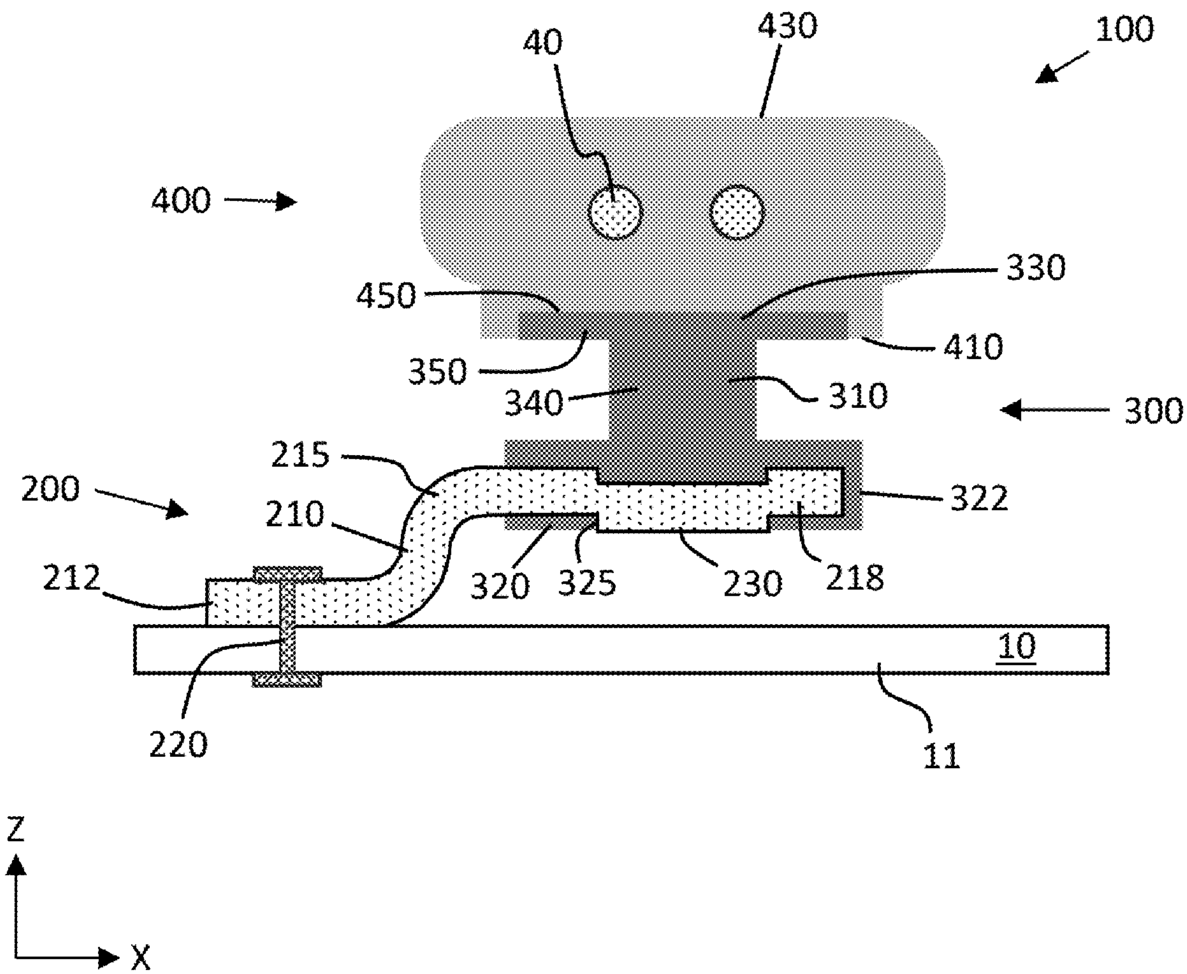
(57) **ABSTRACT**

Electrical interconnections for vehicle, assemblies for electrical interconnections for vehicles, and methods for mounting electrical interconnections to vehicles are provided. An electrical interconnection for a vehicle includes a barbed projection having a proximal end fixed to the vehicle, wherein the projection includes a distal end including outwardly extending barbs; an electrical connector having a selected mounting interface; and a universal connector mount comprising a body formed with a cavity and a corresponding mounting interface, wherein the distal end of the barbed projection is received within the cavity, wherein the barbs pierce the body, and wherein the selected mounting interface of the electrical connector is releasably connected to the corresponding mounting interface of the universal connector mount.



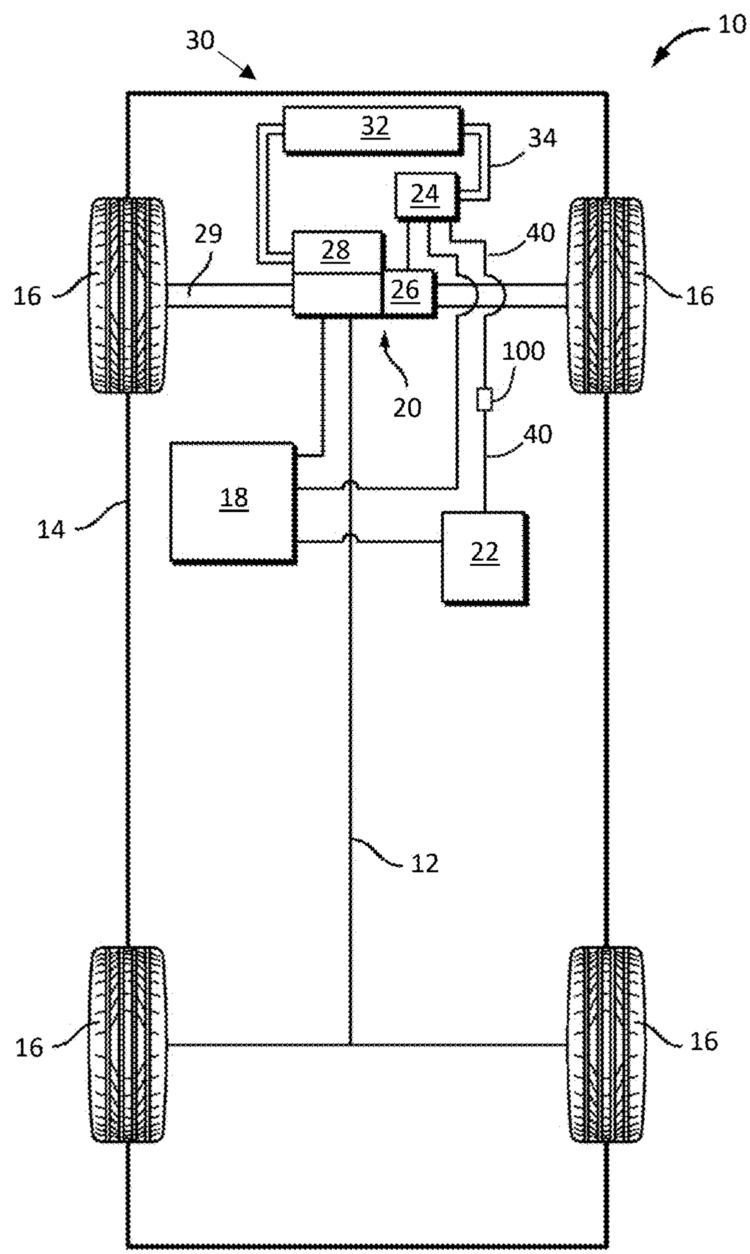
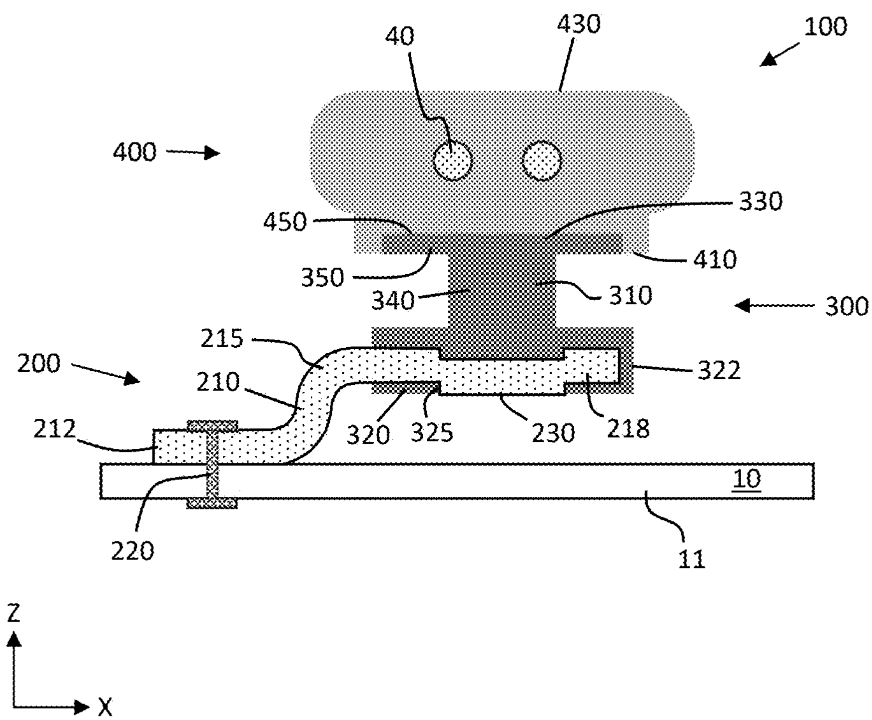


FIG. 1


40
430
100
400
450
330
410
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340
310
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322
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220
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Z
X

FIG. 2

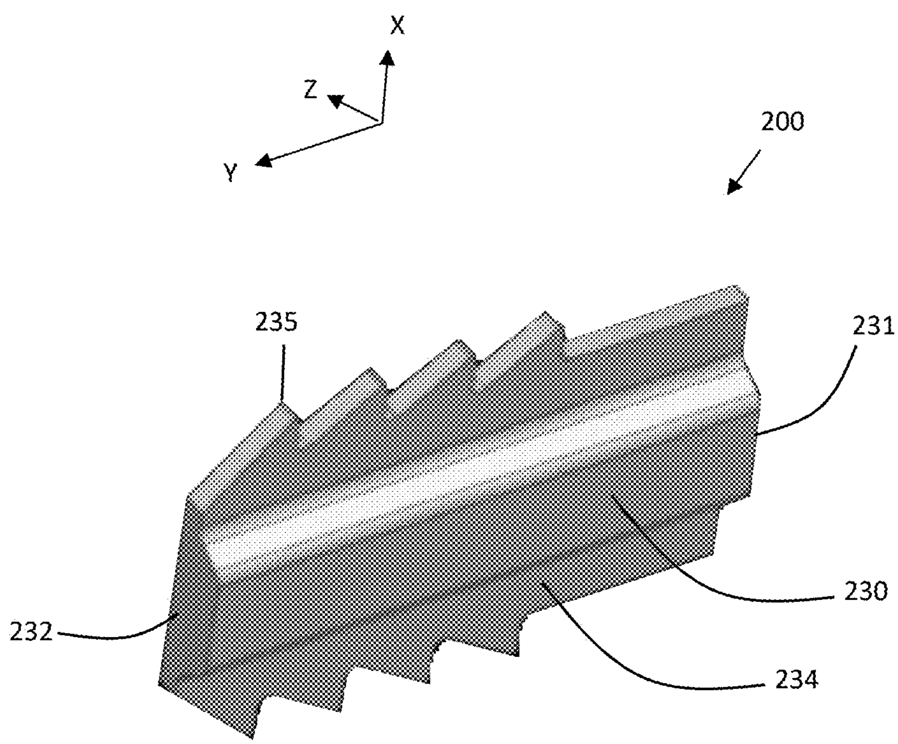


FIG. 3

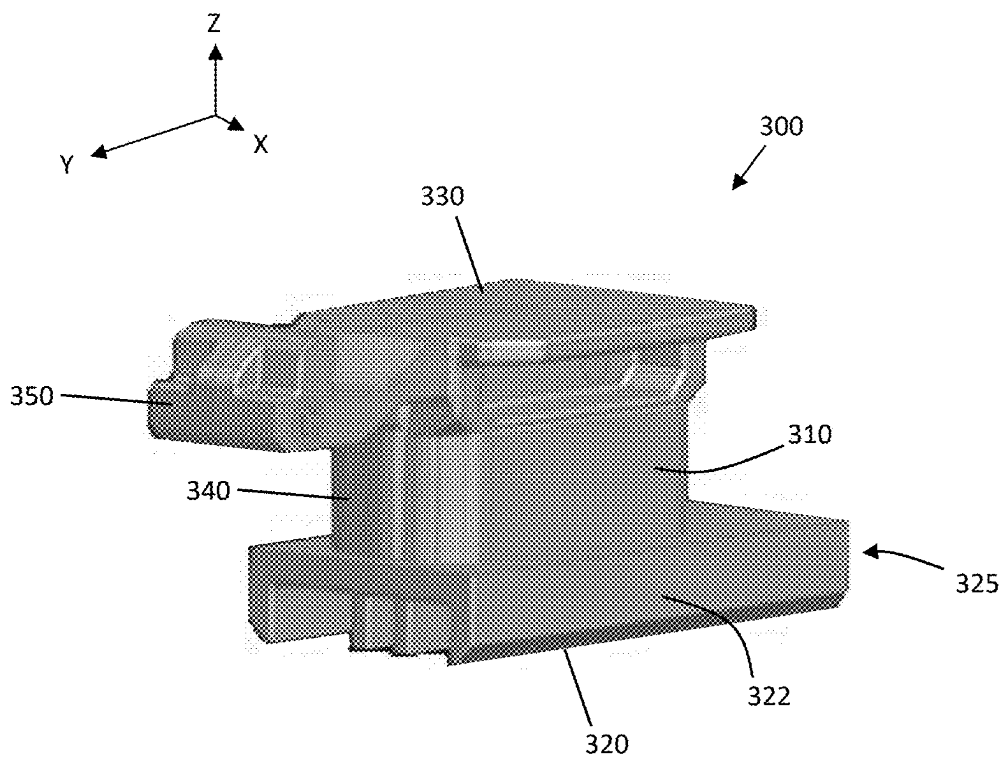


FIG. 4

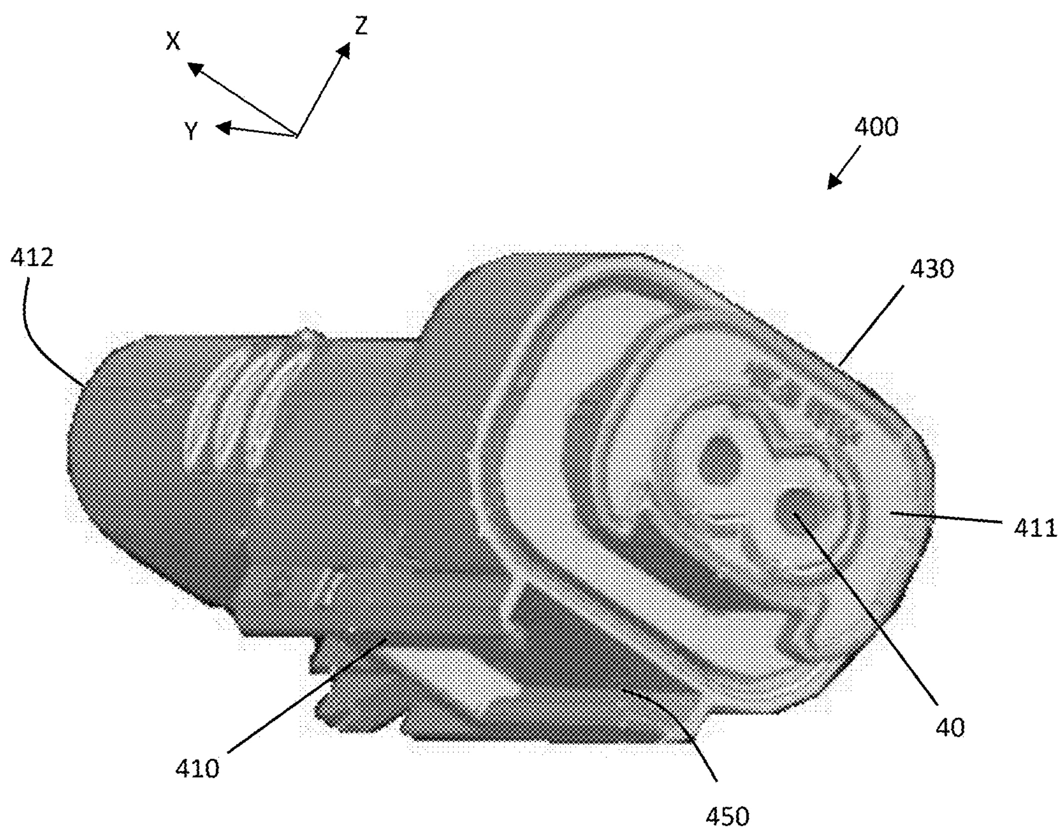


FIG. 5

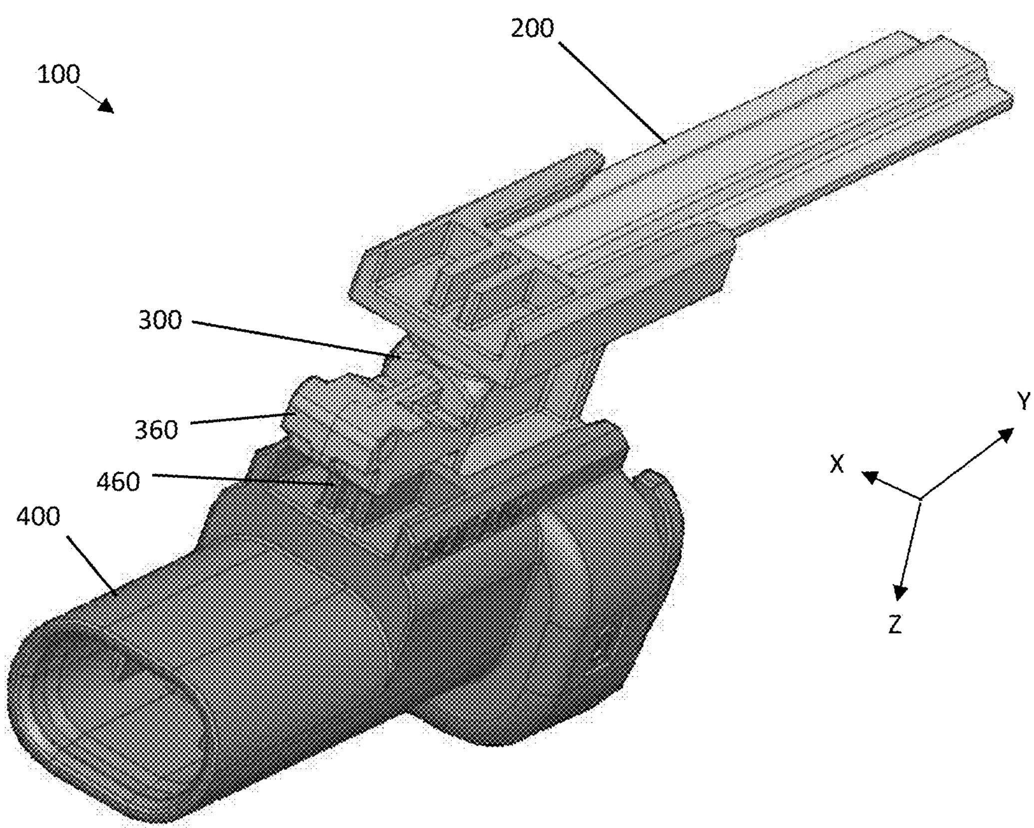


FIG. 6

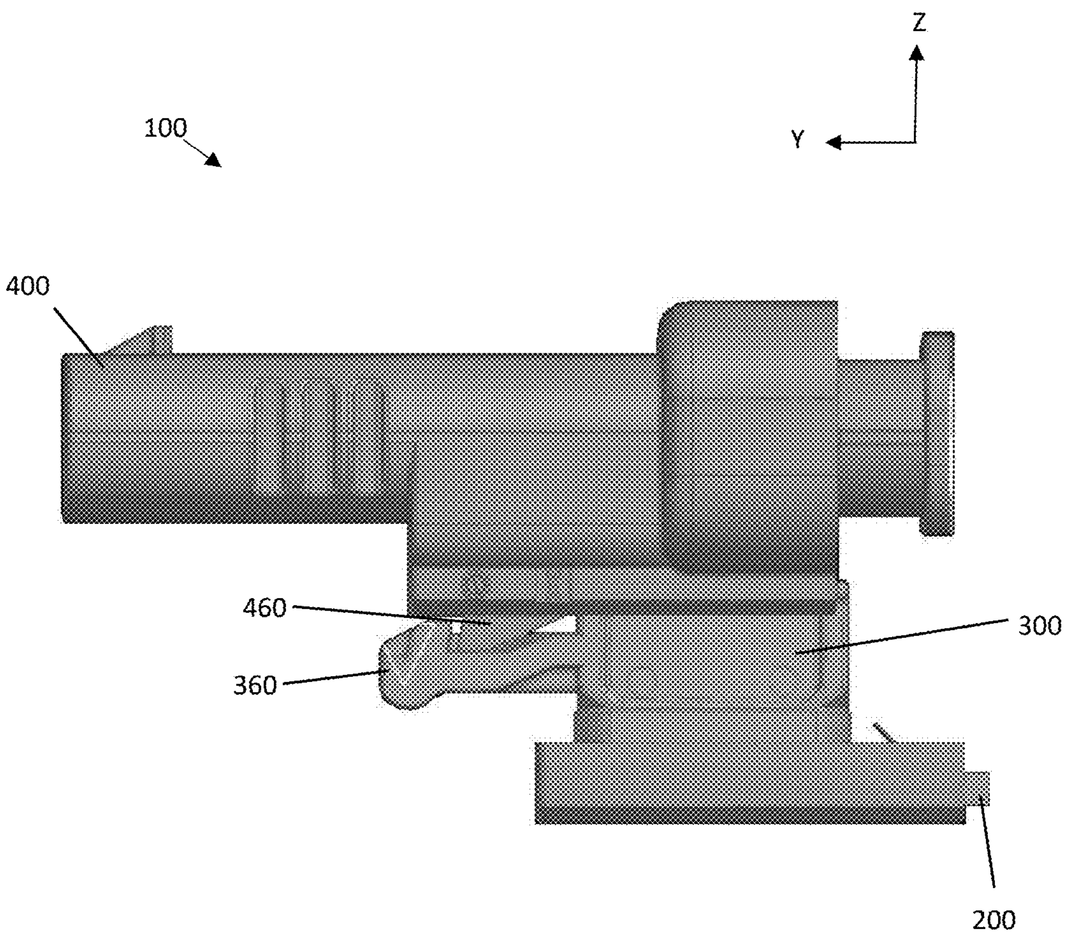


FIG. 7

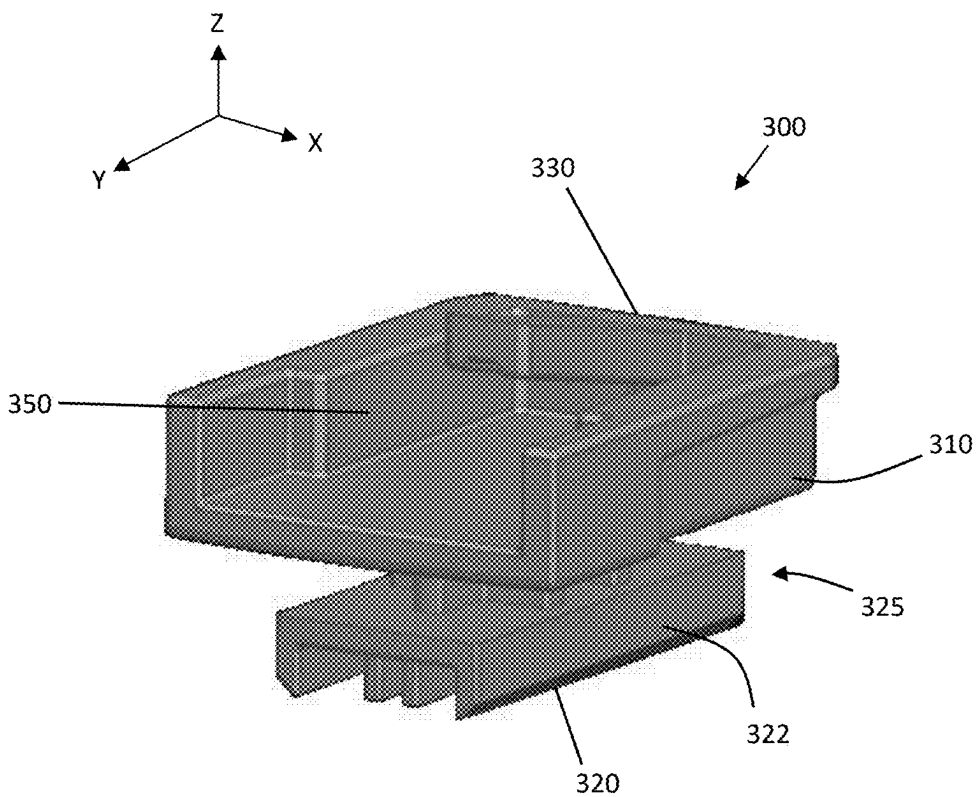


FIG. 8

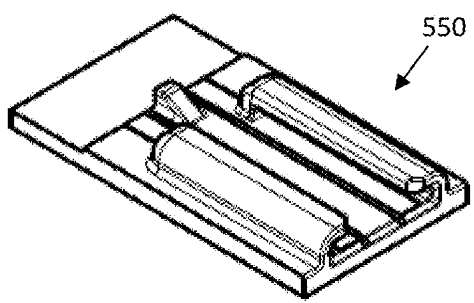


FIG. 9

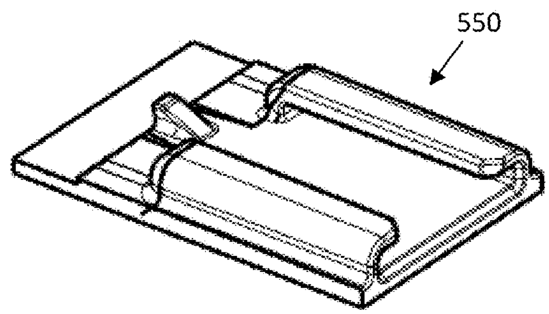


FIG. 10

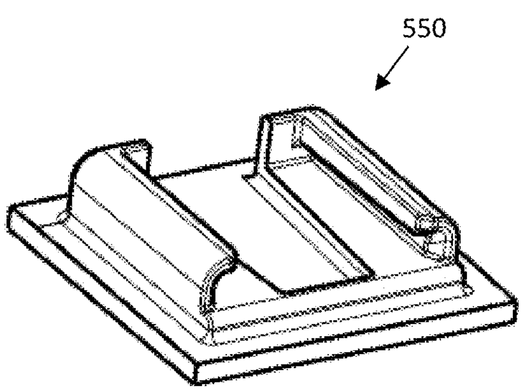


FIG. 11

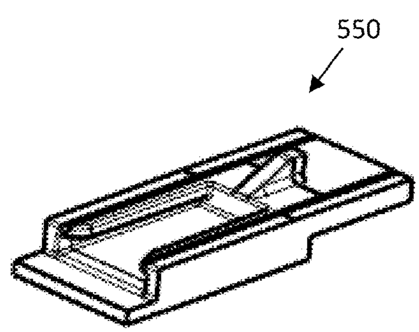
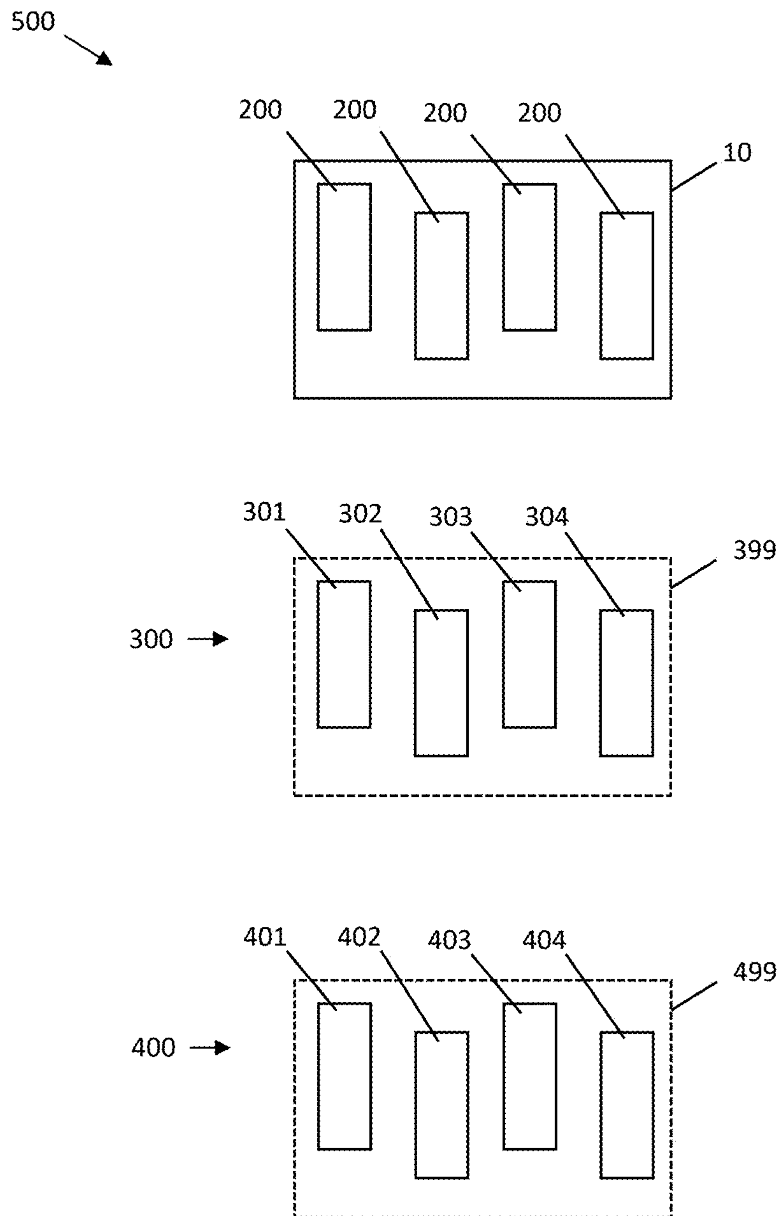


FIG. 12


500
200
200
200
200
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302
303
304
399
300
401
402
403
404
499
400

FIG. 13

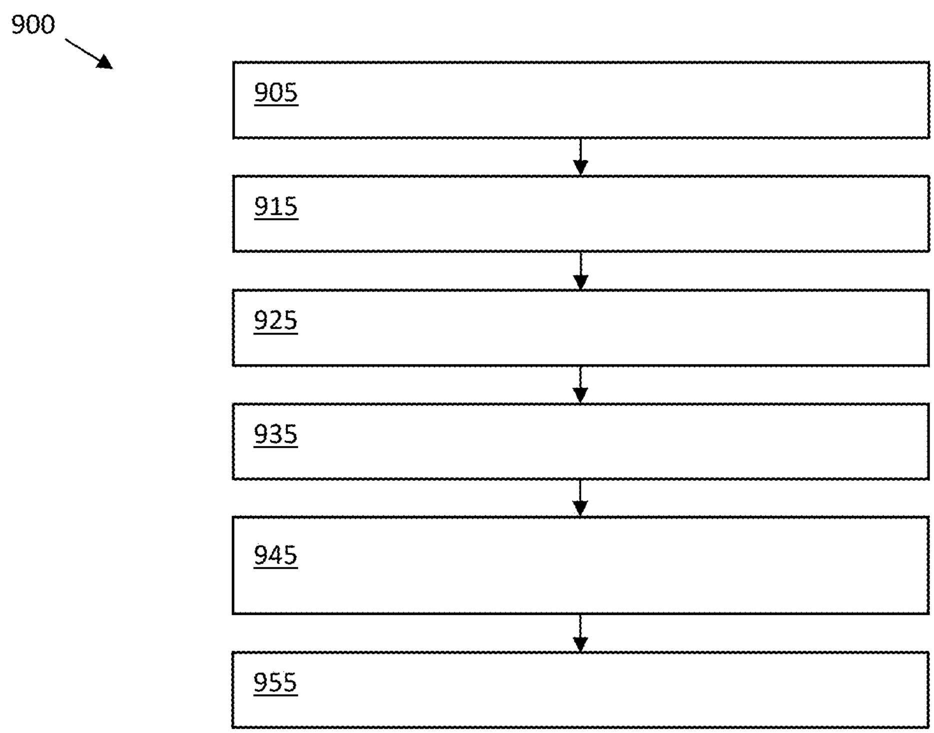

FIG. 14

ELECTRICAL CONNECTOR MOUNTING

INTRODUCTION

[0001] The disclosure relates to electrical connections within vehicles, and may in particular relate to assemblies for ensuring proper mounting of electrical connectors.

[0002] Packaging envelopes for electric circuits supplying electric power to electric machines on vehicles can be limited. Implementation of large capacity electrical systems can be constrained by available packaging space. An example of a large capacity electrical circuit is an electric circuit conducting high voltage electric power between an inverter device and a traction motor. Known design factors of a large capacity electrical circuit include length and routing of high power electric cables based upon electrical energy losses, inductance and electrical noise generation, temperature management and dissipation, and accessibility for assembly and service.

[0003] Accordingly, it is desirable to provide new and improved methods and assemblies for mounting electrical connectors within vehicles. Furthermore, other desirable features and characteristics of the present disclosure will become apparent from the subsequent detailed description and the appended claims, taken in conjunction with the accompanying drawings and the foregoing technical field and background.

SUMMARY

[0004] In an embodiment, an electrical interconnection for a vehicle includes a barbed projection having a proximal end fixed to the vehicle, wherein the projection includes a distal end including outwardly extending barbs; an electrical connector having a selected mounting interface; and a universal connector mount including a body formed with a cavity and a corresponding mounting interface, wherein the distal end of the barbed projection is received within the cavity, wherein the barbs pierce the body, and wherein the selected mounting interface of the electrical connector is releasably connected to the corresponding mounting interface of the universal connector mount.

[0005] In certain embodiments of the electrical interconnection, the selected mounting interface of the electrical connector is selected from a group of regulated mounting interfaces, and the universal connector mount is selected from a group of universal connector mounts, each universal connector mount in the group is formed with the cavity, and each regulated mounting interface is provided on a respective universal connector mount within the group.

[0006] In certain embodiments of the electrical interconnection, the selected mounting interface is a clip slot and the corresponding mounting interface is a retainer clip.

[0007] In certain embodiments of the electrical interconnection, the electrical connector interconnects a first electrical wire and a second electrical wire.

[0008] In certain embodiments of the electrical interconnection, the barbed projection and the universal connector mount are press fit together.

[0009] In certain embodiments of the electrical interconnection, the barbed projection and the universal connector mount are permanently fixed together.

[0010] In certain embodiments of the electrical interconnection, the universal connector mount has a first surface and a second surface opposite the first surface, the cavity is formed on the first surface, and the corresponding mounting interface is formed on the second surface.

[0011] In another embodiment, an assembly for electrical interconnection for a vehicle includes a barbed projection having a proximal end configured for attachment to the vehicle, wherein the projection includes a distal end including outwardly extending barbs; an electrical connector configured to interconnect a first electrical wire and a second electrical wire of the vehicle, wherein the electrical connector has a selected mounting interface; and a lot of universal connector mounts, wherein each universal connector mount in the lot has a body formed with a cavity of a same design configured for receiving the distal end of the barbed projection in a press fit connection, wherein a first universal connector mount in the lot has a first corresponding mounting interface of a first design and a second universal connector mount in the lot has a second corresponding mounting interface of a second design different from the first design, wherein either the first corresponding mounting interface or the second corresponding mounting interface is configured for releasable connection to the selected mounting interface of the electrical connector.

[0012] In certain embodiments of the assembly, the barbs are configured to pierce the body of a selected universal connector mount to provide the press fit connection therebetween.

[0013] In certain embodiments of the assembly, the barbs are configured to pierce the body of a selected universal connector mount to permanently fix the barbed projection and the universal connector mount together.

[0014] In certain embodiments of the assembly, each universal connector mount has a first surface and a second surface opposite the first surface, the cavity is formed on the first surface, and each respective corresponding mounting interface is formed on the second surface.

[0015] In certain embodiments of the assembly, the selected mounting interface of the electrical connector is selected from a group of regulated mounting interfaces, and the universal connector mount is selected from a group of universal connector mounts, each universal connector mount in the group is formed with the cavity, and each regulated mounting interface is provided on a respective universal connector mount within the group.

[0016] In certain embodiments of the assembly, the selected mounting interface is a clip slot and each respective corresponding mounting interface is a retainer clip.

[0017] In certain embodiments of the assembly, a third universal connector mount in the lot has a third corresponding mounting interface of a third design, a fourth universal connector mount in the lot has a fourth corresponding mounting interface of a fourth design different, and the first design, second design, third design, and fourth design are different from one another.

[0018] In another embodiment, a method is provided for mounting an electrical interconnection to a vehicle. The method includes locating a projection on the vehicle; providing an electrical connector having a selected mounting interface; selecting a universal connector mount including a body formed with a cavity and a corresponding mounting interface, wherein the universal connector mount is selected based on a compatibility of the corresponding mounting interface with the selected mounting interface; fixing the universal connector mount to the vehicle by press fitting the projection into the cavity; and releasably connecting the electrical connector to the universal connector mount by engaging the selected mounting interface with the corresponding mounting interface.

[0019] In certain embodiments, the method further includes interconnecting a first electrical wire and a second electrical wire with the electrical connector.

[0020] In certain embodiments of the method, the projection is barbed, and fixing the universal connector mount to the vehicle by press fitting the projection into the cavity includes piercing the body of the universal connector mount with the projection.

[0021] In certain embodiments of the method, the selected mounting interface is a clip slot and the corresponding mounting interface is a retainer clip.

[0022] In certain embodiments of the method, the universal connector mount is selected from a lot of universal connector mounts, in each universal connector mount in the lot the cavity is formed with a same design configured for receiving the projection in a press fit connection, a first universal connector mount in the lot has a first corresponding mounting interface of a first design and a second universal connector mount in the lot has a second corresponding mounting interface of a second design different from the first design, either the first corresponding mounting interface or the second corresponding mounting interface is configured for releasable connection to the selected mounting interface of the electrical connector.

[0023] In certain embodiments of the method, a third universal connector mount in the lot has a third corresponding mounting interface of a third design, a fourth universal connector mount in the lot has a fourth corresponding mounting interface of a fourth design different, and the first design, second design, third design, and fourth design are different from one another.

DESCRIPTION OF THE DRAWINGS

[0024] The present disclosure will hereinafter be described in conjunction with the following drawing figures, wherein like numerals denote like elements, and wherein:

[0025] FIG. **1** is a schematic layout of a vehicle having an electrical interconnection in accordance with exemplary embodiments;

[0026] FIG. **2** is a cross-sectional view schematic of the electrical interconnection of FIG. **1** in accordance with exemplary embodiments;

[0027] FIG. **3** is a perspective view schematic of a projection from the electrical interconnection of FIG. **2** in accordance with exemplary embodiments;

[0028] FIG. **4** is a perspective view schematic of a universal connector mount from the electrical interconnection of FIG. **2** in accordance with exemplary embodiments;

[0029] FIG. **5** is a perspective view schematic of an electrical connector from the electrical interconnection of FIG. **2** in accordance with exemplary embodiments;

[0030] FIG. **6** is a perspective view schematic of the electrical interconnection of FIG. **2** in accordance with exemplary embodiments;

[0031] FIG. **7** is a side view schematic of the electrical interconnection of FIG. **2** in accordance with exemplary embodiments;

[0032] FIG. **8** is a perspective view schematic of another universal connector mount from the electrical interconnection of FIG. **2** in accordance with exemplary embodiments;

[0033] FIGS. **9-12** are perspective view schematics of various mounting interfaces for use in the universal connector mount and/or electrical connector of FIG. **2** in accordance with exemplary embodiments;

[0034] FIG. **13** is a schematic of an assembly for the electrical interconnection of FIG. **2** in accordance with exemplary embodiments; and

[0035] FIG. **14** is a flow chart illustrating a method for mounting an electrical interconnection to a vehicle in accordance with certain embodiments.

DETAILED DESCRIPTION

[0036] The following detailed description is merely exemplary in nature and is not intended to limit the application and uses of embodiments herein. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding introduction, summary or the following detailed description. As used herein, the term "module" refers to any hardware, software, firmware, electronic control unit or component, processing logic, and/or processor device, individually or in any combination, including without limitation: application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality.

[0037] Embodiments of the present disclosure may be described herein in terms of functional and/or logical block components and various processing steps. It should be appreciated that such block components may be realized by any number of hardware, software, and/or firmware components configured to perform the specified functions. For example, an embodiment of the present disclosure may employ various integrated circuit components, e.g., memory elements, digital signal processing elements, logic elements, look-up tables, or the like, which may conduct a variety of functions under the control of one or more microprocessors or other control devices. In addition, those skilled in the art will appreciate that embodiments of the present disclosure may be practiced in conjunction with any number of automated driving systems including cruise control systems, automated driver assistance systems and autonomous driving systems, and that the vehicle system described herein is merely one example embodiment of the present disclosure.

[0038] For the sake of brevity, conventional techniques related to signal processing, data transmission, signaling, control, and other functional aspects of the systems (and the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent example functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in an embodiment of the present disclosure.

[0039] Embodiments herein provide for mounting electrical connectors to vehicles using both a press fit connection and a releasable connection. In certain embodiments, a universal connector mount includes a first end configured to be permanently fixed to a vehicle component such as a barbed projection. The universal connector mount includes a second end that is configured for releasable connection to the electrical connector. The universal connector mount and/or electrical connector may be selected for use with one another.

[0040] With reference to FIG. **1**, certain features of a vehicle **10** are illustrated in functional block diagram form. In certain examples, the vehicle **10** comprises an automobile. In various examples, the vehicle **10** may be any one of a number of different types of automobiles, such as, for example, a sedan, a wagon, a truck, or a sport utility vehicle (SUV), and may be two-wheel drive (2WD) (i.e., rear-wheel drive or front-wheel drive), four-wheel drive (4WD), or all-wheel drive (AWD). The vehicle **10** may also incorporate any one of, or combination of, a number of different types of engines, such as, for example, a gasoline or diesel fueled combustion engine, a "flex fuel vehicle" (FFV) engine (i.e., using a mixture of gasoline and alcohol), a gaseous compound (e.g., hydrogen and/or natural gas) fueled engine, a combustion/electric motor hybrid engine (i.e., such as in a hybrid electric vehicle (HEV)), and an electric motor.

[0041] As depicted in FIG. **1**, the exemplary vehicle **10** generally includes a chassis **12**, a body **14**, four wheels **16**, and an electronic control system **18**. The body **14** is arranged on the chassis **12** and substantially encloses the other components of the vehicle **10**. The body **14** and the chassis **12** may jointly form a frame. The wheels **16** are each rotationally coupled to the chassis **12** near a respective corner of the body **14**.

[0042] In the exemplary embodiment illustrated in FIG. **1**, the vehicle **10** includes an actuator assembly **20**, a battery (or a DC power supply) **22**, and a power converter assembly (e.g., an inverter or inverter assembly) **24**. The actuator assembly **20** may include an electric motor/generator (or motor) **26** and/or a combustion engine **28**.

[0043] Still referring to FIG. **1**, the electric motor **26** and the combustion engine **28** are integrated such that one or both are mechanically coupled to at least some of the wheels **16** through one or more drive shafts **29**.

[0044] The vehicle **10** further includes a thermal management system **30** having a radiator **32** and coolant circulation hoses **34**. The radiator **32** may be connected to the frame at an outer portion thereof. The coolant circulation hoses **34** may provide multiple cooling channels that contain a cooling fluid (i.e., coolant) such as water and/or ethylene glycol (i.e., "antifreeze"). The coolant circulation hoses **34** may be coupled to the combustion engine **28** and/or the inverter **24**. In certain embodiments, these components control the temperature of the battery **22**, inverter **24**, and motor **26** to provide optimal performance.

[0045] The electronic control system **18** is in operable communication with the actuator assembly **20**, the high voltage battery **22**, and the inverter **24**. Although not shown in detail, the electronic control system **18** includes various sensors and automotive control modules, or electronic control units (ECUs), such as an inverter control module, a motor controller, and a vehicle controller, and at least one processor and/or a memory.

[0046] As shown, wires **40** provide for electrical communication between various components. Wires **40** are mounted to the vehicle **10** at various locations by an electrical interconnection **100**.

[0047] Referring to FIG. **2**, an exemplary electrical interconnection **100** is illustrated. As shown, the interconnection **100** may include the vehicle **10**, a projection **200**, an electrical connector **400**, and a universal connector mount **300** interconnecting the projection **200** and the electrical connector **400**.

[0048] In certain embodiments, the projection **200** is located on a bracket **210** that extends laterally in an X-direction from a first end **212** to a second end **218**. At the first end **212**, the bracket **210** is secured to a component **11** of the vehicle **10**. The vehicle component **11** may be a body structure, an engine structure, a door panel, sheet metal, and/or any suitable vehicle location. A fastener **220** such as a bolt may be used to secure the bracket **210** to the vehicle component **11**. In other embodiments, the bracket **210** or projection **200** may be welded or molded to the vehicle component **11**. In certain embodiments, the projection **200** may be part of the vehicle component **11**. An exemplary projection **200** is metal or other rigid material. As shown, the bracket **210** includes a shoulder portion **215** that extends in the Z-direction away from the vehicle component **11**.

[0049] Cross-referencing FIGS. **2** and **3**, at the distal or second end **218** of the bracket **210**, the projection **200** extends longitudinally in a Y-direction perpendicular to the X-direction and to the Y-direction from a proximal end **231** to a distal end **232**. As shown, the projection **200** may include a central rib **230** extending from the proximal end **231** to the distal end **232**. Further, the projection **200** may have wing portions **234** extending away from the rib **230** in the X-direction. As shown, barbs **235** are formed on the wing portions **234** and extend away from the rib **230** in the X-direction to tips.

[0050] Cross-referencing FIGS. **2** and **4**, the universal connector mount **300** includes a body **310** extending in the Z-direction from a bottom surface **320** to a top surface **330**. At the bottom surface **320**, the body **310** includes a platform **322** formed with a cavity **325**. The cavity **325** extends in the Y-direction.

[0051] At the top surface **330**, the universal connector mount **300** includes an engagement or mounting interface or feature **350**. In the illustrated embodiment, the interface **350** is a retainer clip designed in accordance with United States Council for Automotive Research (USCAR) standards, i.e., a USCAR retainer clip.

[0052] As shown, the universal connector mount **300** includes a neck **340** connecting the platform **322** and the engagement interface **350**.

[0053] Cross-referencing FIGS. **2** and **5**, the electrical connector **400** provides for interconnection of wires **40** and may be or include a wire harness. The electrical connector **400** extends in the Y-direction from a first end **411** to a second end **412**. Each end **411** and **412** may receive adjacent wires **40** for electrical interconnection. The electrical connector **400** includes a bottom surface **410** and extends in the Z-direction to a top surface **430**. A mounting interface **450** is formed on or in the bottom surface **410**. In the illustrated embodiment, the mounting interface **450** is an USCAR clip slot or cavity, configured to engage an USCAR retainer clip, such as interface **350** of the universal connector mount **300**.

[0054] While FIGS. **2-5** indicate that an USCAR retainer clip is formed on the universal connector mount **300** and an USCAR clip slot is formed on the electrical connector **400**, it is envisioned that an USCAR retainer clip be formed on the electrical connector **400** and an USCAR clip slot be formed on the universal connector mount **300**. In either case, the universal connector mount **300** and electrical connector **400** are designed and formed for releasable connection.

[0055] FIG. 6 is a perspective view of the electrical interconnection 100 showing the interconnection of the projection 200, universal connector mount 300, and electrical connector 400. FIG. 7 is a side view of the same assembly 100. Cross-referencing FIGS. 2-7, the projection 200 is connected to the universal connector mount 300 by advancing the distal end 232 into the cavity 325 in the Y-direction. The barbs 235 of the projection 200 contact and then pierce the body 310 of the universal connector mount 300. The angled surfaces of the barbs 235 allow for forward movement of the projection 200 into the cavity 325, but prevent withdrawal of the projection 200 from the cavity 325. As a result, the projection 200 may be press fit into the cavity 325 of the universal connector mount 300.

[0056] In certain embodiments, upon press fitting the projection 200 into the cavity 325, the projection 200 and universal connector mount 300 are permanently fixed together. As used herein, the term "permanently fixed" means fixed in a manner such that the universal connector mount 300 and the barbed projection 200 cannot be separated without damaging the universal connector mount 300 and/or, if removed, the universal connector mount 300 cannot be re-fixed to the barbed projection 200.

[0057] On the other hand, the connection between the universal connector mount 300 and the electrical connector 400 is releasable without damage to the components. For example, the interface 350 of the universal connector mount 300 may include a retractable hook 360 and the interface 450 of the electrical connector 400 may include a catch 460 that engage when the interfaces 350 and 450 are engaged, i.e., such as by sliding a retainer clip into a clip slot. The interfaces 350 and 450 may be disengaged by pressing the hook 360 away from engagement with the catch 460 and sliding the universal connector mount 300 and the electrical connector 400 away from one another. Other connection, engagement, and release features may be used.

[0058] For example, FIG. 8 illustrates an alternative design of a universal connector mount 300 having a different interface 350.

[0059] FIGS. 4 and 8 illustrate two embodiments of interfaces 350 that comply with the United States Council for Automotive Research (USCAR) standards. It is contemplated herein that the interfaces 350 and 450 may be designed in compliance with any of the approved USCAR standard clip slot and retainer clip designs.

[0060] United States Council for Automotive Research (USCAR) provides for several standardized clip slot and retainer clip designs. For example, USCAR standards include the 11 mm clip slot 550 of FIG. 9, the 17 mm clip slot 550 of FIG. 10, the 23 mm clip slot 550 of FIG. 11, and the 7 mm clip slot 550 of FIG. 12. Any of these or other designs may be used to provide releasable connection between the universal connector mount 300 and electrical connector 400.

[0061] The embodiments of FIGS. 1-8 allow for releasably mounting electrical connectors 400 to a vehicle 10 while still utilizing press fit, permanent, connections to the vehicle 10. The universal connector mount 300 is formed with a cavity for receiving and holding the barbed projection 200 permanently. Further, the universal connector mount 300 is formed with a clip slot or cavity with dimensions for receiving the retainer clip of the electrical connector 400. Alternatively, the electrical connector 400 is formed with a clip slot or cavity with dimensions for receiving the retainer clip of the universal connector mount 300.

[0062] In certain embodiments, the electrical interconnection 100 for vehicle 10 includes a barbed projection 200 having a proximal end 231 fixed to the vehicle 10. The projection 200 includes a distal end 232 including outwardly extending barbs 235.

[0063] The electrical interconnection 100 further includes an electrical connector 400 configured to interconnect first and second electrical wires 40. The electrical connector 400 has a selected mounting interface 450.

[0064] The electrical interconnection 100 further includes a universal connector mount 300 comprising a body 310 formed with a cavity 325 and a corresponding mounting interface 350, wherein the distal end 232 of the barbed projection 200 is received within the cavity 325, wherein the barbs 235 pierce the body 310, and wherein the selected mounting interface 450 of the electrical connector 400 is releasably connected to the corresponding mounting interface 350 of the universal connector mount 300.

[0065] In certain embodiments, the electrical interconnection 100 for vehicle 10 is provided as an assembly 500. For example, the assembly 500 may include each component 200, 300, and 400 before interconnection. Each barbed projection 200 may have a proximal end 231 configured for attachment to the vehicle 10 and a distal end 232 including outwardly extending barbs 235.

[0066] Further, a plurality of components 300 and 400 may be provided in the assembly 500. For example, a lot 399 of universal connector mounts 300 may be provided and a lot 499 of electrical connectors 400 may be provided.

[0067] In the lot 499 of electrical connectors 400, each electrical connector 400 is configured to interconnect electrical wires of the vehicle 10, Each electrical connector 400 has a selected mounting interface 450. Each selected mounting interface 450 of the electrical connectors 400 is selected from a group of regulated mounting interfaces, such as USCAR interfaces. For example, a first electrical connector 401 in the lot 499 has a first selected mounting interface 450 of a first design, a second electrical connector 402 in the lot 499 has a second selected mounting interface 450 of a second design, a third electrical connector 403 in the lot 499 has a third selected mounting interface 450 of a third design, a fourth electrical connector 404 in the lot 499 has a fourth selected mounting interface 450 of a fourth design. The selected mounting interfaces 450 of the electrical connectors 401, 402, 403, and 404 are different from one another. For example, each of the selected mounting interfaces 450 may be one of the clip slots 550 of FIGS. 9-12 or may correspond and mate with one of the clip slots 550 of FIGS. 9-12, i.e., be a retainer clip.

[0068] In the lot 399 of universal connector mounts 300, each universal connector mount 300 has a body 310 formed with a cavity 325 of a same design configured for receiving the distal end of the barbed projection 200 in a press fit connection. A first universal connector mount 301 in the lot 399 has a first corresponding mounting interface 350 of a first design, a second universal connector mount 302 in the lot 399 has a second corresponding mounting interface 350 of a second design different from the first design, a third universal connector mount 303 in the lot 399 has a third corresponding mounting interface 350 of a third design different from the first design and second design, and a fourth universal connector mount 304 in the lot 399 has a fourth corresponding mounting interface **350** of a fourth design different from the first design, second design, and third design. In exemplary embodiments, the corresponding mounting interface of one of the universal connector mount **301**, **302**, **303**, and **304** is configured for releasable connection to the selected mounting interface **450** of the selected electrical connector **400**. For example, each of the corresponding mounting interfaces **350** may be one of the clip slots **550** of FIGS. **9-12** or may correspond and mate with one of the clip slots **550** of FIGS. **9-12**, i.e., be a retainer clip.

[0069] Referring now to FIG. **14**, a flow chart of a method **900** for mounting an electrical interconnection to a vehicle is provided. As shown, method **900** includes, at operation **905**, locating a projection **200** on the vehicle **10**. Locating the projection **200** on the vehicle **10** may include forming the projection **200** and/or mounting the projection **200** on the vehicle **10**.

[0070] Further, method **900** may include, at operation **915**, providing an electrical connector **400** having a selected mounting interface **450**. In certain embodiments, the electrical connector **400** may be selected from a lot **499** of electrical connectors of various designs. In other embodiments, the electrical connector **400** may be pre-selected, and/or may be present in the vehicle **10** and installed to interconnect wires **40** therein.

[0071] Method **900** may continue at operation **925** with selecting a universal connector mount **300** with a corresponding mounting interface **350**. Selection of the universal connector mount **300** may be based on a compatibility of the corresponding mounting interface **350** with the selected mounting interface **450**. In certain embodiments, the universal connector mount **300** is selected from a lot **399** of universal connector mounts **300**.

[0072] As shown, method **900** may include fixing the universal connector mount **300** to the vehicle **10** at operation **935**. Operation **935** may include press fitting the projection **200** into the cavity **325**. Press fitting the projection **200** into the cavity **325** may include piercing the body **310** of the universal connector mount **300** with the projection **200**.

[0073] Further, method **900** may include at operation **945** releasably connecting the electrical connector **400** to the universal connector mount **300**. For example, operation **945** may include engaging the selected mounting interface **450** with the corresponding mounting interface **350**.

[0074] Method **900** may also include interconnecting a first electrical wire **40** and a second electrical wire **40** with the electrical connector **400** at operation **955**. As noted above, in certain embodiments, the wires **40** may already be interconnected at the electrical connector **400**.

[0075] While FIG. **14** illustrates a sequential order of operations, it is noted that the method **900** is not limited to such a sequential order and that the operations may be performed in any suitable logical order.

[0076] While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the disclosure in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the exemplary embodiment or exemplary embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope of the disclosure as set forth in the appended claims and the legal equivalents thereof.

What is claimed is:

1. An electrical interconnection for a vehicle comprising:

a barbed projection having a proximal end fixed to the vehicle, wherein the projection includes a distal end including outwardly extending barbs;

an electrical connector having a selected mounting interface; and

a universal connector mount comprising a body formed with a cavity and a corresponding mounting interface, wherein the distal end of the barbed projection is received within the cavity, wherein the barbs pierce the body, and wherein the selected mounting interface of the electrical connector is releasably connected to the corresponding mounting interface of the universal connector mount.

2. The electrical interconnection of claim **1**, wherein the selected mounting interface of the electrical connector is selected from a group of regulated mounting interfaces, and wherein the universal connector mount is selected from a group of universal connector mounts, wherein each universal connector mount in the group is formed with the cavity, and wherein each regulated mounting interface is provided on a respective universal connector mount within the group.

3. The electrical interconnection of claim **1**, wherein the selected mounting interface is a clip slot and the corresponding mounting interface is a retainer clip.

4. The electrical interconnection of claim **1**, wherein the electrical connector interconnects a first electrical wire and a second electrical wire.

5. The electrical interconnection of claim **1**, wherein the barbed projection and the universal connector mount are press fit together.

6. The electrical interconnection of claim **1**, wherein the barbed projection and the universal connector mount are permanently fixed together.

7. The electrical interconnection of claim **1**, wherein the universal connector mount has a first surface and a second surface opposite the first surface, wherein the cavity is formed on the first surface, and wherein the corresponding mounting interface is formed on the second surface.

8. An assembly for electrical interconnection for a vehicle comprising:

a barbed projection having a proximal end configured for attachment to the vehicle, wherein the projection includes a distal end including outwardly extending barbs;

an electrical connector configured to interconnect a first electrical wire and a second electrical wire of the vehicle, wherein the electrical connector has a selected mounting interface; and

a lot of universal connector mounts, wherein each universal connector mount in the lot has a body formed with a cavity of a same design configured for receiving the distal end of the barbed projection in a press fit connection, wherein a first universal connector mount in the lot has a first corresponding mounting interface of a first design and a second universal connector mount in the lot has a second corresponding mounting interface of a second design different from the first design, wherein either the first corresponding mounting interface or the second corresponding mounting interface is configured for releasable connection to the selected mounting interface of the electrical connector.

9. The assembly of claim **8**, wherein the barbs are configured to pierce the body of a selected universal connector mount to provide the press fit connection therebetween.

10. The assembly of claim **8**, wherein the barbs are configured to pierce the body of a selected universal connector mount to permanently fix the barbed projection and the universal connector mount together.

11. The assembly of claim **8**, wherein each universal connector mount has a first surface and a second surface opposite the first surface, wherein the cavity is formed on the first surface, and wherein each respective corresponding mounting interface is formed on the second surface.

12. The assembly of claim **8**, wherein the selected mounting interface of the electrical connector is selected from a group of regulated mounting interfaces, and wherein the universal connector mount is selected from a group of universal connector mounts, wherein each universal connector mount in the group is formed with the cavity, and wherein each regulated mounting interface is provided on a respective universal connector mount within the group.

13. The assembly of claim **8**, wherein the selected mounting interface is a clip slot and each respective corresponding mounting interface is a retainer clip.

14. The assembly of claim **8**, wherein a third universal connector mount in the lot has a third corresponding mounting interface of a third design, wherein a fourth universal connector mount in the lot has a fourth corresponding mounting interface of a fourth design different, and wherein the first design, second design, third design, and fourth design are different from one another.

15. A method for mounting an electrical interconnection to a vehicle, the method comprising:

locating a projection on the vehicle;

providing an electrical connector having a selected mounting interface;

selecting a universal connector mount comprising a body formed with a cavity and a corresponding mounting interface, wherein the universal connector mount is selected based on a compatibility of the corresponding mounting interface with the selected mounting interface;

fixing the universal connector mount to the vehicle by press fitting the projection into the cavity; and

releasably connecting the electrical connector to the universal connector mount by engaging the selected mounting interface with the corresponding mounting interface.

16. The method of claim **15**, further comprising interconnecting a first electrical wire and a second electrical wire with the electrical connector.

17. The method of claim **15**, wherein the projection is barbed, and wherein fixing the universal connector mount to the vehicle by press fitting the projection into the cavity comprises piercing the body of the universal connector mount with the projection.

18. The method of claim **15**, wherein the selected mounting interface is a clip slot and the corresponding mounting interface is a retainer clip.

19. The method of claim **15**, wherein the universal connector mount is selected from a lot of universal connector mounts, wherein in each universal connector mount in the lot the cavity is formed with a same design configured for receiving the projection in a press fit connection, wherein a first universal connector mount in the lot has a first corresponding mounting interface of a first design and a second universal connector mount in the lot has a second corresponding mounting interface of a second design different from the first design, wherein either the first corresponding mounting interface or the second corresponding mounting interface is configured for releasable connection to the selected mounting interface of the electrical connector.

20. The method of claim **19**, wherein a third universal connector mount in the lot has a third corresponding mounting interface of a third design, wherein a fourth universal connector mount in the lot has a fourth corresponding mounting interface of a fourth design different, and wherein the first design, second design, third design, and fourth design are different from one another.

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